

Networking in the Cloud Era





COURSE STRUCTURE

Day 1: Ethernet Switching, VLANs, Routing, OSPF & EIGRP

Day 2: STP, EtherChannel, HSRP, DHCP, NAT & Data Center Architecture

Day 3: Networking in the Cloud Era — AWS VPC, EC2 & Security

Day 4: Cloud Network Services & Hybrid Connectivity

Day 5: High Availability, Troubleshooting & Network Management Platforms

AGENDA FOR DAY 3

1. **Networking in the Cloud Era**
2. Security & Connectivity
3. Infrastructure as Code (IaC) Concepts



NETWORKING IN THE CLOUD ERA

CLOUD TECHNOLOGIES

- The cloud is on-demand access to computing resources (servers, storage, databases, networking, software) over the Internet.
- Instead of owning physical infrastructure, organizations rent resources from cloud providers.
- Examples of the top 3 cloud providers



Azure



Google Cloud

CLOUD SERVICES MODELS

- IaaS (Infrastructure as a Service): Virtual servers and storage
- PaaS (Platform as a Service): Tools for app development
- SaaS (Software as a Service): Ready-to-use apps

WHY COMPANIES ARE MOVING TO THE CLOUD

- Cost Efficiency – Pay for what you use; no need to buy hardware.
- Scalability – Easily handle peak demand and scale down when idle.
- Accessibility – Work from anywhere with an internet connection.
- Innovation Speed – Faster deployment of new features and services.
- Resilience & Security – Built-in backups, redundancy, and global security standards.

REAL WORLD USE CASES

- Netflix: Uses AWS for global content delivery and scaling. Precursor of Chaos Engineering
- Airbnb: Runs its platform on cloud infrastructure.
- Banks & Insurance Companies: Using private/hybrid cloud for secure data processing.

MAPPING TRADITIONAL TO AWS NETWORKING

On-Prem Concept	AWS Equivalent
VLAN / Switch	Subnet
Router	Route Table + Gateways
Firewall	Security Group / Network ACL
IP Address Mgmt	VPC CIDR Block
DHCP Server	AWS DHCP Options Set
Physical Cable	AWS Virtual Network Fabric

AWS GLOBAL NETWORK STRUCTURE

- VPC CIDR Block: Defines IP range (e.g., 10.0.0.0/16)
- Subnets: Divide your network into public and private zones, 1 AZ per subnet
- Route Tables: Direct traffic between subnets or to external destinations
- Gateways: Enable communication to the internet or other networks
- Elastic Network Interfaces (ENIs): Virtual NICs attached to resource

POP QUIZ:

What is a key reason organizations move to the cloud for cost efficiency?

- A. They pay for hardware in advance
- B. They pay only for what they use
- C. They have fixed monthly costs
- D. They must maintain servers on-site



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Which cloud model allows developers to build and deploy applications without managing servers?

- A. IaaS
- B. PaaS
- C. SaaS
- D. DaaS



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LAB 00: AWS EC2 INTRODUCTION

- Goal: Launch and manage AWS EC2 instances.
- Steps:
 - Create AWS account and configure billing alerts.
 - Launch EC2 instance with proper security groups.
 - Connect via SSH and configure basic services.
 - Understand AWS pricing and cost optimization.

SECURITY & CONNECTIVITY

PUBLIC VS PRIVATE SUBNETS

- Similar to physical network, we will use public and private subnets for security

Subnet Type	Typical Resources
Public	Web servers, bastions
Private	Databases, internal services

ROUTE TABLES AND ROUTING

- Local Route: Enables intra-VPC communication between subnets within the same VPC.
- Default Route (0.0.0.0/0): Directs internet-bound traffic to an Internet Gateway (IGW) for public subnets or to a NAT Gateway (NAT GW) for private subnets.
- Custom Routes:
 - Used for advanced network connectivity, such as
 - VPC Peering connections
 - Transit Gateway attachments (covered in later modules)
- Subnet Association:
 - Each subnet must be explicitly associated with a route table to define its traffic flow.

SECURITY GROUPS (SGS)

- Stateful: Return traffic is automatically allowed when the inbound rule permits it.
- Attachment: Applied directly to Elastic Network Interfaces (ENIs) or EC2 instances.
- Rule Type: Allow rules only — all other traffic is implicitly denied.
- Example:
 - Allow HTTP (port 80) from anywhere
 - Allow SSH (port 22) only from an admin IP

NETWORK ACLS (NACLs)

- Stateless:
 - Both inbound and outbound rules must be explicitly defined — return traffic is not automatically allowed.
- Scope:
 - Applied at the subnet level, affecting all resources within that subnet.
- Rule Evaluation:
 - Processed in ascending order based on rule number (lowest number takes priority).
- Purpose:
 - Provides an additional layer of security beyond Security Groups.

SG VS NACL COMPARISON

Feature	Security Group
Scope	Instance (ENI)
Stateful?	
Default	Deny all inbound / allow all outbound
Rule Type	Allow only
Use Case	App tier filtering

POP QUIZ:

What is a public subnet in AWS typically used for?

- A. Hosting backend databases
- B. Hosting internet-facing services
- C. Internal-only services
- D. Private DNS management



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What type of firewall are AWS Security Groups?

- A. Stateless
- B. Stateful
- C. Dynamic
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POP QUIZ:

How are NACLs different from Security Groups?

- A. NACLs are stateful
- B. NACLs are stateless and evaluated per subnet
- C. NACLs apply only to EC2 instances
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LAB 1 OVERVIEW: BUILD YOUR FIRST VPC

- Goal: Create secure cloud network infrastructure.
- Steps:
 - Design VPC with public and private subnets.
 - Configure Internet Gateway and NAT Gateway.
 - Set up route tables for traffic flow.
 - Launch instances in appropriate subnets.

LAB 02: AWS SECURITY GROUPS

- Goal: Implement instance-level firewall rules.
- Steps:
 - Create security groups with specific rules.
 - Configure inbound and outbound traffic policies.
 - Test connectivity between security group members.
 - Apply least-privilege security principles.

LAB 03: AWS NETWORK ACLS

- Goal: Configure subnet-level network filtering.
- Steps:
 - Create custom Network ACL rules.
 - Compare stateful vs stateless filtering.
 - Test NACL rule precedence and numbering.
 - Troubleshoot connectivity with dual security layers.

INFRASTRUCTURE AS CODE (IAC) CONCEPTS

WHY IAC MATTERS

- Traditional Approach:
Manual configuration is slow, error-prone, and leads to inconsistencies between environments.
- Infrastructure as Code (IaC): Infrastructure is defined, managed, and provisioned through code instead of manual processes.
- Key Benefits:
 - Version Control: Track and manage changes like application code.
 - Repeatability: Easily reproduce environments across stages.
 - Automation: Faster, consistent deployments with reduced human error.

TOOLS TO ENABLE IAC

- Multiple tools will allow us to achieve IaC

Tool	Focus
Terraform	Multi-cloud
CloudFormation	AWS native
Ansible	Configuration
Chef/Puppet	OS-level mgmt